

RA'ED ALASMAR

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PROFESSIONAL SUMMARY

Electrical Engineering student specializing in intelligent communication systems, with hands-on experience in RAN implementation and optimization through internships at **Zain Jordan**—working directly with **Nokia** RAN equipment. Completed the full **CCNA** track with expertise in BTS deployment, KPI-driven optimization, RF planning, and OSS-based performance analysis. Currently conducting AI-based anomaly detection research and developing a RAG-based faculty chatbot as part of the **Google Developer Group** (GDG) on Campus BME, combining deep networking knowledge with practical ML and software development experience.

WORK EXPERIENCE

C Programming Lab Instructor - Budapest Univ. of Tech. & Econ. 09/2025 - 01/2026

- Led weekly labs for ~25 students, teaching C programming from fundamentals to real-life projects.
- Mentored students through algorithms, data structures, and system-level problem solving.

RAN Optimization Intern, Zain Jordan 07/2025 - 09/2025

- Analyzed LTE/5G KPIs including handover success, call drops, RSRP/RSRQ/SINR, and throughput.
- Used OSS tools and RTWP/spectrum scans to detect uplink interference and optimize coverage.
- Supported frequency planning, BSIC checks, and handover optimization with cross-functional teams.

RAN Implementation Intern, Zain Jordan 06/2025 - 07/2025

- Assisted in full RAN site implementation: BTS setup, ERP ordering, and microwave link deployment.
- Validated BBU, RU, IDU/ODU, and RF jumpers; monitored KPIs and alarms via OSS systems.

PROJECTS

Enterprise Office Building Infrastructure

- Designed and configured a VLAN-segmented enterprise network with OSPF, DHCP, inter-VLAN routing, SSH access, and MAC-based port security in **Cisco Packet Tracer**.

Software-Defined Network Traffic Guardian (Python/Linux)

- Built a Python-based SDN controller using Ryu to manage OpenFlow switches in Mininet.
- Implemented automated traffic rerouting logic to resolve network congestion in real-time.

Task Manager (C++)

- Engineered a CLI task manager with dynamic memory management, task prioritization, and deadline tracking.

VIK Faculty Chatbot (RAG-based, Ongoing)

- Built an AI chatbot for BME VIK faculty using RAG, FastAPI, Gemini, and ChromaDB — achieving **97%**.

AI-Based Anomaly Detection Research (Ongoing)

- Replacing traditional ensemble voting with fingerprint-driven AutoML routing to dynamically select the best model per time series pattern.
- Optimizing ensemble weights and detection thresholds to improve accuracy and generalization across diverse anomaly types.

EDUCATION

Bachelor of Electrical Engineering 09/2022 - 01/2027

Budapest University of Technology and Economics

- Received the Stipendium Hungaricum full scholarship award (top **10%** acceptance rate).
- Tech Team Member, Google Developer Group (GDG) on Campus BME.
- Member of BME Sollertia, bridging international and Hungarian students.

CERTIFICATES

- Cisco Networking Academy** - CCNA (Full Track): 140+ labs, 50+ hours.
- Google** - Introduction to Git and GitHub
- DeepLearning.AI** - Machine Learning Specialization

SKILLS

Telecommunications: UMTS/LTE/NR Architecture, RAN Optimization, KPI Analysis, Nokia RAN Equipment, OSS Systems.

Networking: Router/Switch Configuration, OSPF, EIGRP, VLANs, DHCP, NAT, Subnetting, ACLs.

Programming: Python, C, C++, SQL.

AI & ML: RAG Frameworks, Anomaly Detection, AutoML (TPOT), Ensemble Methods, LLM Integration.

Tools: Git & GitHub, Jira, MS Office, Packet Tracer, Linux CLI.